

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Winter Examination – 2024

Course: B.TECH.

Branch : MECHANICAL ENGINEERING

Semester : III

Subject Code & Name: BTMEC302_Y19 MATERIALS SCIENCE AND METALLURGY

Max Marks: 60

Date: 07/02/2025

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	Line Defects are also known as		1
	a. Stacking Fault b. Point Defect c. Volume Defect. d. Dislocations		
2	Rise in stress during plastic deformation is called		1
	a. Toughness b. Hardness c. work Hardening d. Stress Relieving		
3	Hardness test may give the information about		1
	a. wear Resistance b. Machinability c. Effectiveness of the heat treatment d. All of these		
4	The failure that happen due to cyclic loading is called....		1
	a. Creep Failure b. Fatigue Failure c. Shears Failure d. Tensile Failure		
5	Phase diagram is obtained by		1
	a. Very Fast Cooling b. Very Fast Heating c. Very Slow Cooling d. Initially Cooling and then Heating		
6	According to the Hume Ruther which factor does not govern solid solubility?		1
	a. Crystal Structure b. Relative Size c. Relative Density d. Relative Valency		
7	Which of the following constituents of steels is softest and least strong.....		1
	a. Austenite b. Pearlite c. Cementite d. Ferrite		
8	Which type of test is mostly used for detecting surface defects in crane hooks		1
	a. Magnaflux b. Ultrasonic c. Radiography d. None of these		

9	Water proof Emery papers are used for					1
	a. Dry Polishing	b. Wet Polishing	c. Fine Polishing	d. Electrolytic Polishing		
10	Powder Metallurgy is most suitable method for ----- Hardened Materials.					1
	a. Dispersion	b. Age	c. Strain	d. Precipitation		
11	Quenching is not required in -----					1
	a. Carburizing	b. Nitriding	c. Carbonitriding	d. Hardening		
12	Which one may be used as a material for Transducer in Ultrasonic Test?					1
	a. Steel	b. Quartz	c. Polymer	d. None of these		
Q. 2 Solve the following.						12
A) Derive the Expressions for planar atomic on (100), (110) and (111) planes of both BCC and FCC crystal structures.						6
B) Discuss Induction hardening process w.r.t. principle of working process details advantages and limitations.						6
Q. 3 Solve the following.						12
A) Explain dispersion strengthening w.r.t. basic mechanisms, critical factors advantages and commercial examples.						6
B) Explain the principle of working of optical metallurgical Microscope. Compare it with electron microscope.						6
Q. 4 Solve Any Two of the following.						12
A) Describe the procedure of specimen preparation of microscopy.						6
B) What is the importance of T-T-T diagrams? Explain the procedure to determine these diagrams with the help of schematic diagrams						6
C) Explain the principle of magnetic particle testing (magna flux) and what are the methods of magnetization? state advantages and disadvantages of demagnetization						6
Q. 5 Solve Any Two of the following.						12
A) Define heat treatment and give its objective. Give the names of two different heat treatment processes along with the major objective of each.						6
B) Draw Fe-Fe ₃ equilibrium diagram and explain various phases and critical temperatures present in it.						6
C) Discuss Sulphur print test w.r.t. purposes, significance, procedure, and chemical reactions.						6

Q. 6	Solve Any Two of the following.		12
A)	Describe ultrasonic inspection technique w.r.t principle of working types, and applications		6
B)	Discuss Vickers Hardness tests w.r.t. principle of working, indenter details, formula, advantages and limitations.		6
C)	Explain Slip Mechanism of Plastic Deformation w.r.t. conceptual meaning its occurrence due to the movement of edge and screw dislocations and comparison with twinning.		6
*** End ***			